

3.1 $B_1 = (\vec{u}_1, \dots, \vec{u}_m)$, $B_2 = (\vec{v}_1, \dots, \vec{v}_m)$ dvě báze $\mathbb{R}^n$. Označme G_j Gramovu matici posoup. vzhledem ke sř. sk. s. jako G_j (pro $j = 1, 2$)

Ukažte, že následující podmínky jsou ekvivalentní:

(a) existuje ob. matice Q řádu n t.j. $Q\vec{u}_i = \vec{v}_i$ pro každé $i \in \{1, \dots, m\}$

(b) $G_1 = G_2$

$$G_1 = \begin{pmatrix} \vec{u}_1 \cdot \vec{u}_1 & \vec{u}_1 \cdot \vec{u}_2 & \vec{u}_1 \cdot \vec{u}_3 & \dots & \vec{u}_1 \cdot \vec{u}_m \\ \vec{u}_2 \cdot \vec{u}_1 & \vec{u}_2 \cdot \vec{u}_2 & \vec{u}_2 \cdot \vec{u}_3 & \dots & \vec{u}_2 \cdot \vec{u}_m \\ \vec{u}_3 \cdot \vec{u}_1 & \vec{u}_3 \cdot \vec{u}_2 & \vec{u}_3 \cdot \vec{u}_3 & \dots & \vec{u}_3 \cdot \vec{u}_m \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \vec{u}_m \cdot \vec{u}_1 & \vec{u}_m \cdot \vec{u}_2 & \vec{u}_m \cdot \vec{u}_3 & \dots & \vec{u}_m \cdot \vec{u}_m \end{pmatrix}; \quad G_2 = \begin{pmatrix} \vec{v}_1 \cdot \vec{v}_1 & \vec{v}_1 \cdot \vec{v}_2 & \vec{v}_1 \cdot \vec{v}_3 & \dots & \vec{v}_1 \cdot \vec{v}_m \\ \vec{v}_2 \cdot \vec{v}_1 & \vec{v}_2 \cdot \vec{v}_2 & \vec{v}_2 \cdot \vec{v}_3 & \dots & \vec{v}_2 \cdot \vec{v}_m \\ \vec{v}_3 \cdot \vec{v}_1 & \vec{v}_3 \cdot \vec{v}_2 & \vec{v}_3 \cdot \vec{v}_3 & \dots & \vec{v}_3 \cdot \vec{v}_m \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \vec{v}_m \cdot \vec{v}_1 & \vec{v}_m \cdot \vec{v}_2 & \vec{v}_m \cdot \vec{v}_3 & \dots & \vec{v}_m \cdot \vec{v}_m \end{pmatrix}$$

Q je ob. matice, $Q = (\vec{q}_1 | \vec{q}_2 | \dots | \vec{q}_m)$ pak $\vec{q}_i \perp \vec{q}_j \quad \forall i, j \in \{1, \dots, m\} \quad i \neq j$

Platí $Q\vec{u}_i = \vec{v}_i$, tedy $\sum_{j=1}^m q_{ji} \vec{u}_j = \vec{v}_i \Rightarrow \vec{q}_i \cdot \vec{q}_j = 0$

Bud' A_j matice řádu n t.j. $A_1 = (\vec{u}_1 | \vec{u}_2 | \dots | \vec{u}_m)$, $A_2 = (\vec{v}_1 | \vec{v}_2 | \dots | \vec{v}_m)$

Pak $G_1 = A_1^T A_1$ a $G_2 = A_2^T A_2$

pokud $\vec{u}_i = \begin{pmatrix} u_{1i} \\ u_{2i} \\ u_{3i} \\ \vdots \\ u_{ni} \end{pmatrix} \quad u_{ji} \in \{1, \dots, n\}$

a $\vec{v}_i = \begin{pmatrix} v_{1i} \\ v_{2i} \\ v_{3i} \\ \vdots \\ v_{ni} \end{pmatrix}$

pokud $q_{ji} = \begin{pmatrix} q_{1i} \\ q_{2i} \\ q_{3i} \\ \vdots \\ q_{mi} \end{pmatrix} \quad q_{ji} \in \{1, \dots, m\}$

$$A_1 = \begin{pmatrix} u_{11} & u_{12} & \dots & u_{1m} \\ u_{21} & u_{22} & & \\ u_{31} & u_{32} & \dots & \\ \vdots & \vdots & \ddots & \vdots \\ u_{m1} & u_{m2} & \dots & u_{mm} \end{pmatrix}_{m \times n}$$

$$A_1^T = \begin{pmatrix} u_{11} & u_{21} & u_{31} & \dots & u_{m1} \\ u_{12} & u_{22} & u_{32} & \dots & u_{m2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ u_{1m} & u_{2m} & u_{3m} & \dots & u_{mm} \end{pmatrix}_{n \times m}$$

$Q\vec{u}_j = \begin{pmatrix} \sum_{i=1}^m q_{1i} u_{ij} \\ \sum_{i=1}^m q_{2i} u_{ij} \\ \vdots \\ \sum_{i=1}^m q_{mi} u_{ij} \end{pmatrix}$... můžeme říct, že je také ob.?

$Q^* = Q^{-1}$

$$Q = \begin{pmatrix} q_{11} & q_{12} & q_{13} & q_{14} & \dots & q_{1m} \\ q_{21} & q_{22} & & & & \\ q_{31} & & & & & \\ \vdots & & & & & \\ q_{m1} & & & & & \end{pmatrix}$$

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$$Q \vec{u}_i = \vec{v}_i$$

$$Q^{-1} Q \vec{u}_i = Q^{-1} \vec{v}_i$$

$$\vec{u}_i = Q^{-1} \vec{v}_i$$

$$\left(\begin{matrix} q_{11} & \dots & q_{1n} \\ q_{21} & & q_{2n} \\ \vdots & & \vdots \\ q_{m1} & & q_{mn} \end{matrix} \right) \begin{pmatrix} v_{1i} \\ v_{2i} \\ \vdots \\ v_{mi} \end{pmatrix} = \begin{pmatrix} \vec{q}_1 \cdot \vec{v}_i \\ \vec{q}_2 \cdot \vec{v}_i \\ \vdots \\ \vec{q}_m \cdot \vec{v}_i \end{pmatrix}$$

$\Rightarrow$ pro i -tý vektor báze $B_1 = (u_1, \dots, u_n)$ je roven std. sk. součinu sloupců matice

$Q = (\vec{q}_1 | \vec{q}_2 | \dots | \vec{q}_n)$ s i -tým vektorem báze $B_2 = (\vec{v}_1, \dots, \vec{v}_n)$

$$\vec{u}_1 = \begin{pmatrix} \vec{q}_1 \cdot \vec{v}_1 \\ \vec{q}_2 \cdot \vec{v}_1 \\ \vdots \\ \vec{q}_n \cdot \vec{v}_1 \end{pmatrix}, \quad \vec{u}_2 = \begin{pmatrix} \vec{q}_1 \cdot \vec{v}_2 \\ \vec{q}_2 \cdot \vec{v}_2 \\ \vdots \\ \vec{q}_n \cdot \vec{v}_2 \end{pmatrix} \dots$$

$$Q^* \vec{v}_1 \quad Q^* \vec{v}_2$$

poskládáme součiny do $G_1 \Rightarrow$

$$\begin{pmatrix} Q^* \vec{v}_1 \cdot Q^* \vec{v}_1 & Q^* \vec{v}_1 \cdot Q^* \vec{v}_2 & Q^* \vec{v}_1 \cdot Q^* \vec{v}_3 & \dots & Q^* \vec{v}_1 \cdot Q^* \vec{v}_n \\ Q^* \vec{v}_2 \cdot Q^* \vec{v}_1 & Q^* \vec{v}_2 \cdot Q^* \vec{v}_2 & & & \\ Q^* \vec{v}_3 \cdot Q^* \vec{v}_1 & & & & \\ \vdots & & & & \\ Q^* \vec{v}_n \cdot Q^* \vec{v}_1 & & & & Q^* \vec{v}_n \cdot Q^* \vec{v}_n \end{pmatrix}$$

std. sk. součin nad $\mathbb{C}/\mathbb{R}$

$$\vec{u} \cdot \vec{v} = \vec{u}^* \cdot \vec{v}$$

$$= (\vec{u}_1 \ \vec{u}_2 \ \vec{u}_3 \ \vec{u}_4 \dots \vec{u}_n) \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}$$

pro $\vec{u} = \begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix}, \vec{v} = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}$

$$\vec{u}, \vec{v} \in V$$

$$= G_2 = (\vec{v}_i \cdot \vec{v}_j) \quad (a) \Rightarrow (b) \quad \checkmark$$

použijeme na součiny $Q^* \vec{v}_i \cdot Q^* \vec{v}_j$

$$= (Q^* \vec{v}_i)^* \cdot (Q^* \vec{v}_j) = \vec{v}_i Q^{**} Q^* \vec{v}_j = \vec{v}_i \underbrace{Q Q^*}_{I_n} \vec{v}_j = \vec{v}_i \cdot \vec{v}_j, \text{ pro } i, j \in \{1, \dots, n\}$$